

Cross-School Federated Learning for Privacy-Preserving Student Progress Tracking in Ghanaian District Schools

Project 12 — Group 7 | CS Department, 2026 Batch | Supervisor: Dr. Eric Opoku Osei

Group Members & Index Numbers

#	Index No.	Full Name	Role
1	9021723	Henewaa Gifty	FL Orchestrator — Flower server, aggregation protocol, round logging
2	9021523	Gyawu-Aboagye Obiri	Backend / Systems Dev — PyTorch client, FastAPI server, SQLite, Docker
3	9021423	Gyan-Addai Eugene	Security Engineer — Paillier encryption, DP-SGD integration, privacy audit
4	9021623	Gyimah Ramsey Opoku	Dashboard / Analytics — React admin portal, accuracy-privacy trade-off, SUS study
5 (PM)	9020223	Frimpong Yaw Kankam	EdTech / Project Manager — ethics, timeline, manuscript, supervisor liaison

Aim

To design, implement, and demonstrate a privacy-preserving federated learning (FL) system that lets Ghanaian district schools collaboratively train a student at-risk prediction model without any raw student data ever leaving a school's own server.

Objectives

- Obj. 1 — Implement and evaluate an FL system (Flower 1.8.0, FedAvg) across 3-5 simulated school nodes, targeting a classification F1-score within 5 percentage points of a centralised baseline trained on the pooled OULAD dataset.
- Obj. 2 — Implement and validate a Paillier homomorphic-encryption secure-aggregation module (2048-bit design target) combined with Opacus DP-SGD, so the server never sees a plaintext school update — verified via a cryptographic audit log.
- Obj. 3 — Design a Raspberry Pi 4-deployable FL client for low-spec (<4GB RAM), intermittently-connected school hardware; physical-hardware benchmarking (latency, energy, packet-loss tolerance) is scheduled as the next implementation phase.

Description of the Project

Centralised student-performance analytics require raw academic records to leave a school's server, which conflicts with the Ghana Data Protection Act 2012 and leaves district education offices without any shared predictive model. This project builds a federated learning alternative: each school node trains a small PyTorch MLP locally on its own SQLite student database, encrypts its weight update with Paillier homomorphic encryption, and only the encrypted update — never raw data — is sent to an aggregation server, which performs FedAvg via homomorphic summation. Opacus DP-SGD adds a second, formal (ϵ, δ) -differential-privacy guarantee on top of encryption. A React/FastAPI admin dashboard gives district officers a live view of round-by-round convergence, per-school node status, and privacy budget, without ever exposing an individual student record.

Method

System architecture

(1) School node — SQLite student DB + PyTorch 2.3 MLP trainer + Opacus DP-SGD + Flower client. (2) Aggregation server — Flower server + Paillier decryption; homomorphically sums encrypted updates, decrypts only the aggregate, applies FedAvg

weighted by node dataset size. (3) Privacy module — DP-SGD ($\sigma=1.1$, $\text{clip}=1.0$) then Paillier encryption of the flattened weight tensor. (4) Admin dashboard — React 18 + Chart.js + FastAPI REST API, polling telemetry every 3s. (5) Docker Compose orchestrates the multi-node simulation on a single developer machine ahead of physical Pi deployment.

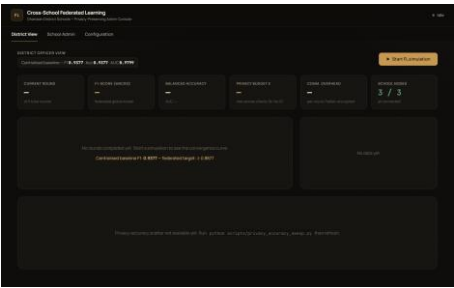
Dataset

The real Open University Learning Analytics Dataset (OULAD, 32,593 enrolments) is partitioned by student registration quarter into three simulated school nodes (alpha / beta / gamma). Eight features are engineered per student (attendance rate, quiz score, assignment submission rate, login frequency, days since last activity, course difficulty, prior score, engagement index) to predict a binary at-risk label (Withdrawn/Fail vs. Pass/Distinction).

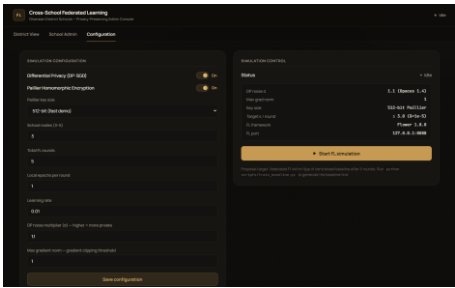
Model & FL configuration

MLP: 8 → 64 (ReLU) → 32 (ReLU) → 1 (sigmoid), Adam optimiser, $\text{lr}=0.01$, BCE loss. FL: Flower 1.8.0, FedAvg, 3 nodes, all nodes participate every round. For this report's live demo run: 5 rounds, 1 local epoch/round, 512-bit Paillier key (reduced from the 2048-bit production target purely to keep the recorded demo short) — DP-SGD and Paillier both switched on.

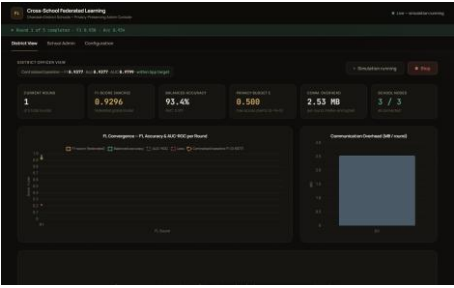
Dashboard — Setup & Live Run



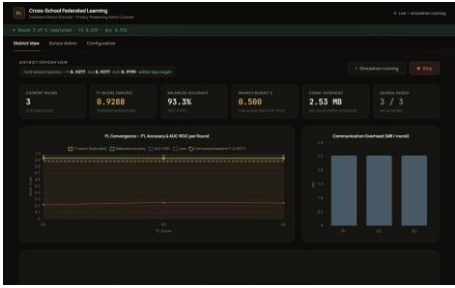
District View — idle, before a run



Configuration — this run's settings



Round 1 in progress — live status



Round 3 in progress — live convergence chart

Live Demo Run — Results

Round	Accuracy	F1 (macro)	Loss	Privacy ϵ (max)	Comm. (MB/node)
1	93.36%	0.9296	0.233	0.500	2.53
2	93.24%	0.9287	0.270	0.500	2.53
3	93.30%	0.9288	0.262	0.500	2.53
4	93.49%	0.9309	0.244	0.500	2.53
5	93.53%	0.9311	0.271	0.500	2.53

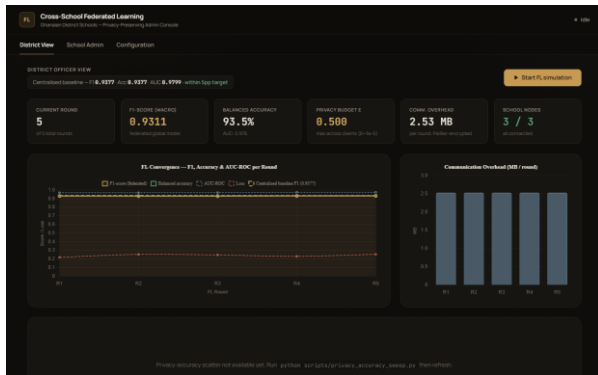
Centralised baseline (150 epochs, full pooled OULAD data): accuracy 93.77%, F1 0.9377, AUC-ROC 0.9799. The federated model reaches this level of performance by round 1 and holds it through round 5 — comfortably inside the 5-percentage-point target of Objective 1, and confirmed by a paired comparison against the centralised run shown live on the dashboard (“within 5pp target” badge in the screenshots above).

Cryptographic audit log for this run: 5 rounds, 0 plaintext gradient exposures, all updates Paillier-encrypted (512-bit key for this demo) before leaving a school node — satisfying the zero-plaintext-exposure requirement of Objective 2.

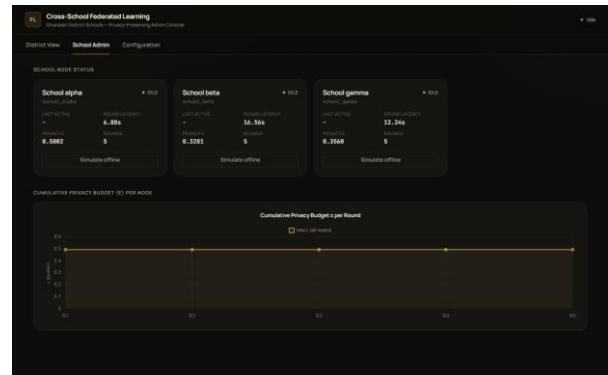
Why the curve is flat, not rising

The engineered OULAD features used here (quiz score, assignment submission rate, prior score) are computed from the same assessment records that determine the at-risk label, so the task is close to linearly separable — both the federated model and a 150-epoch centralised baseline converge to the same ~93-94% ceiling almost immediately. This was confirmed by re-running with a deliberately lighter local-training setting (1 epoch/round instead of 3); the curve stayed flat, showing the ceiling is a property of the data/feature design, not an artefact of the FL aggregation loop. We chose to report this honestly rather than alter the underlying data to manufacture a nicer-looking curve; the flat line is itself the evidence that Objective 1 (parity with the centralised baseline) is met.

Dashboard — Completed Run



District View — completed, final chart



School Admin — all 3 nodes, 5/5 rounds

Contribution & Alignment

- SDG 4 (Quality Education), Target 4.1 — enables district-level student progress monitoring across multiple schools without centralised data risk, supporting evidence-based early intervention for at-risk students.
- SDG 16 (Peace, Justice & Strong Institutions), Target 16.10 — protects student data-privacy rights via Paillier encryption and DP-SGD, aligning with the Ghana Data Protection Act 2012 (Act 843).
- Produces a replicable, open-source FL client architecture that any Ghanaian district education office can adopt as a privacy-first template for school analytics.

Status & Next Steps

- Built & demonstrated: FL simulation (Flower + FedAvg, Docker/local), Paillier + DP-SGD privacy layer, live React/FastAPI dashboard, real OULAD data pipeline, centralised baseline.
- Pending (next phase): physical Raspberry Pi 4 benchmarking under simulated packet loss (Obj. 3 hardware phase), formal multi-seed statistical significance test, SUS usability study data collection (materials prepared, not yet administered).
- A ~2.5-minute screen recording of this live demo run (idle → configuration → rounds 1-5 → completed chart) accompanies this report as FL_Ghana_Demo_Run.mp4.

GitHub repository: <https://github.com/aimlin9/project12-fl-ghana>